Cover Letter

November 13, 2026

Data and AI Hiring Team
BNSF Railway | BNSF Tech
Fort Worth, Texas

Re: Data Scientist I/II (Remote - US), Job ID 95561

Dear Hiring Team:

I am interested in the Data Scientist I/II opportunity at BNSF Railway. I hold a PhD in Electrical and Computer Engineering from the University of Oklahoma and bring a combination of applied data science, machine-learning engineering, cloud data workflows, and research on operationally complex systems. I am particularly interested in BNSF Tech's focus on using reliable data and predictive analytics to support transportation operations and practical business decisions.

In my current machine-learning engineering assignment supporting AT&T's E911 service-assurance operations, I work with operational event data to develop classification, diagnostic, and model-validation workflows. One root-cause classification effort improved automated classification performance from approximately 20% to 80%. The work requires careful handling of noisy and heterogeneous records, class imbalance, distribution shifts, error analysis, and collaboration with network and software teams to turn technical results into useful operational insight.

Previously, at Impac Exploration Services, I built Python- and Azure-based ingestion, transformation, and data-validation workflows using Databricks, Azure Functions, and Synapse Analytics. These processes reduced data-preparation time by 40% and improved the reliability of analytical outputs by 20%. My AT&T Labs research also involved analyzing more than 500,000 wireless diagnostic logs and applying machine-learning methods to detect anomalies and reduce diagnostic time by 20%. These experiences are relevant to BNSF's needs in complex-data analysis, predictive modeling, pipeline integrity, troubleshooting, and stakeholder communication.

My technical foundation includes Python, SQL, pandas, NumPy, scikit-learn, PyTorch, LightGBM, Spark/PySpark, Azure Data Factory, Azure DevOps, cross-validation, statistical analysis, and reproducible data-science experimentation. Although my professional experience is in telecommunications and exploration data rather than freight rail, I have repeatedly adapted analytical approaches to unfamiliar operational domains. I would bring that same collaborative, evidence-based approach to asset reliability, process improvement, and data-driven decisions in rail transportation.

I would welcome the opportunity to contribute to BNSF Tech's Data and AI team and learn the operational priorities of the freight network. Thank you for considering my qualifications.

Sincerely,

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